

# GuardShield AI: Real-Time LLM Guardrail Proxy

## B.Tech CSE (AIML) Final Year Project Proposal | 7th & 8th Semester Roadmap

**Executive Summary:** GuardShield AI is a high-performance, real-time security sidecar proxy designed to sit between user applications and Large Language Models (LLMs). It intercepts user inputs to block prompt injection attacks and scans streaming LLM output tokens to detect hallucinations and policy violations with sub-25ms latency.

### 1. Problem Statement & Need

As enterprise adoption of GenAI accelerates, two major vulnerabilities threaten LLM deployments: **(1) Prompt Injections & Adversarial Jailbreaks** that trick models into revealing system secrets or breaking safety rules, and **(2) Model Hallucinations** where LLMs generate plausible-sounding but factually wrong statements. GuardShield AI provides an automated, low-latency defense layer to solve both problems.

### 2. System Architecture & Workflow

- 1. Pre-Inference Inspection:** Scans incoming user prompts using fine-tuned DeBERTa-v3 models for DAN patterns, prompt leakage, and jailbreak attempts.
- 2. Local LLM Execution:** Routes verified prompts to a high-speed local LLM engine (Ollama Llama-3.1 / Qwen-2.5).
- 3. Post-Inference Streaming Inspection:** Calculates token probability logit entropy and verifies facts against context using NLI entailment scoring.
- 4. Telemetry Dashboard:** Logs security events, attack vectors, and hallucination rates in real-time.

### 3. Semester Breakdown (Minor vs. Major Project)

| Phase           | Core Deliverables & Modules                                                                                                                                                                                                                                            | Key Output                                        |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| 7th Sem (Minor) | <ul style="list-style-type: none"><li>Jailbreak &amp; Prompt Injection Classifier (DeBERTa-v3)</li><li>Token Entropy &amp; NLI Hallucination Detector</li><li>Model Benchmark Evaluation on HaluEval &amp; JailbreakBench</li></ul>                                    | Working Model Pipeline & Benchmark Report         |
| 8th Sem (Major) | <ul style="list-style-type: none"><li>FastAPI / Rust Async SSE Streaming Sidecar Proxy</li><li>Autonomous LLM Red-Teaming Adversarial Attack Agent</li><li>Live Security Telemetry Dashboard (React/Streamlit)</li><li>(Bonus) iOS App Prompt Injection Demo</li></ul> | Production Proxy + Live UI + Research Paper Draft |

### 4. Recommended Tech Stack & Setup

- Local LLM Inference:** Ollama (Running llama3.1:8b / qwen2.5:14b locally)
- Deep Learning Core:** PyTorch with Apple Silicon MPS (Metal Performance Shaders) acceleration
- Proxy Microservice:** FastAPI (Async SSE Streaming) / Docker
- Security Classifier Models:** Hugging Face Transformers, DeBERTa-v3, ONNX Runtime
- Frontend & Analytics:** Streamlit / React + TailwindCSS
- Zero Hardware Cost:** 100% software-based, runs locally on developer laptops without cloud API costs.

## 5. Team & Career Impact

**Why this project sets your resume apart:** Generative AI Safety, Hallucination Reduction, and Red-Teaming are top priority hiring areas in enterprise AI. Publishing an open-source GitHub repository and submitting a research paper to a Scopus-indexed student conference will make your team stand out to recruiters.